

3(a)	The first law of thermodynamics states that the increase in internal energy of a system is the sum of the heat supplied to the system and the work done on the system.	[2] or zero
3(b)(i)	$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$ $\frac{(1.04 \times 10^5)(2.90 \times 10^{-4})}{314} = \frac{p_2(2.90 \times 10^{-5})}{790}$ $p_2 = 2.62 \times 10^6 \text{ Pa}$	[1] correct substitution [1] answer
3(b)(ii)	$\Delta U = Q + W$ $= 0 + 90$ $= 90 \text{ J}$	[1] answer
3(b)(iii)	Internal energy of an ideal gas U is the total random kinetic energy of the gas molecules. The random kinetic energy of the gas molecules is dependent on temperature $T \Rightarrow U \propto T$. Since there is an increase in internal energy of the neon gas, this indicates that there is a rise in the temperature of the gas during compression.	[1] U = total random KE [1] $KE \propto T \Rightarrow U \propto T$ mention increase in internal energy
3(c)(i)	Initial internal energy $U_i = \frac{3}{2}(3.00)(8.31)(400) + \frac{3}{2}(1.00)(8.31)(600) = 22437 \text{ J}$ Final internal energy $U_f = \frac{3}{2}(4.00)(8.31)(T)$ $\frac{3}{2}(4.00)(8.31)(T) = 22437$ $T = 450 \text{ K}$	[1] value of U_i [1] answer
3(c)(ii)	$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3RT}{M_m}}$ $\sqrt{\langle c^2 \rangle} \propto \frac{1}{\sqrt{M_m}}$ For the same temperature for both gases: $\frac{\sqrt{\langle c^2 \rangle}_{He}}{v} = \frac{\sqrt{(M_m)_{Ne}}}{\sqrt{(M_m)_{He}}} = \sqrt{\frac{20}{4}}$ $\sqrt{\langle c^2 \rangle}_{He} = 2.24 v$	[1] correct substitution [1] answer